

PulseRank

Marketplace Ranking & Evaluation Platform

Failure Mode: Is this ranking model actually better, or is the evaluation lying?

IPS Debiasing

MMR Diversity

Seller Gini

Offline A/B

Delayed Attribution

0.134 -> 0.522

NDCG@10 (IPS corrected)

+289%

Evaluation lift

HOLD_SIMULATED

A/B decision

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1 The Problem

User: Marketplace Operators and Growth Teams

The primary users of PulseRank are marketplace operators and growth teams at e-commerce platforms — the people responsible for ensuring that ranking models are actually working, that sellers are getting fair visibility, and that evaluation frameworks are telling the truth.

Marketplace ranking teams were evaluating their rankers using click data that was systematically biased by display position — an item shown at rank 1 receives 8x the clicks of the same item at rank 10 regardless of quality. Models trained and evaluated on this data looked 4x worse than they actually were. The consequence: good rankers were being held back because the evaluation framework charged the display layer's decisions to the model. Meanwhile, pure relevance ranking was starving long-tail sellers of visibility, concentrating impressions on bestsellers and reducing catalog diversity — a problem that eventually surfaces as seller churn.

Why This Problem Is Hard

- **Biased training and evaluation.** The position-click curve is not a model failure — it is an artifact of how the display layer allocates attention. Items at rank 1 receive roughly 8x the clicks of items at rank 10 regardless of intrinsic quality. A model trained on this data learns position patterns, not relevance patterns.
- **The evaluation framework lies.** Naive NDCG computed on position-biased click labels penalizes models for decisions the display layer made. The same ranker scores $\text{NDCG@10} = 0.134$ on biased labels and 0.522 after IPS correction — a 4x difference that reflects evaluation corruption, not model quality.
- **Good models get withheld.** Teams operating on biased metrics hold back rankers that would actually improve the marketplace. The cost is not measured in model quality — it is measured in delayed launches, seller revenue foregone, and competitive position.
- **Long-tail seller starvation.** Pure relevance ranking concentrates impressions on bestsellers. Long-tail sellers — who often carry unique or specialized inventory — receive systematically less visibility than their catalog quality warrants. Over time, this surfaces as seller churn and catalog contraction.
- **Conversion attribution is noisy.** Marketplace purchase intent does not resolve at click time. Naive last-click attribution misattributes conversions that follow a multi-session decision cycle, corrupting the signal used for ranker evaluation.

The Core Evaluation Problem

The position-bias trap is a specific case of a general problem: when training data and evaluation data are both corrupted by the same upstream decision process (display ordering), the corruption is invisible to standard metrics. You cannot detect the bias from within the biased evaluation framework. You need an external correction — IPS — that models the data-generating process and unweights the position advantage.

2 The Product Decisions

What Was Built

Component	Decision	Rationale
IPS position-bias correction	Click $\times$ (1/propensity(rank)), clip max_weight=5.0	Treats position bias as missing-data problem; weights control variance
Hybrid candidate generation	Popularity scorer + content-based scorer, merged top-100	Popularity gives stable coverage; content-based improves tail recall
MMR diversity reranking	lambda=0.70, max_per_seller=2, category_cap=50%	Prevents impression concentration; balances relevance vs diversity
Seller Gini governance	Gini coefficient constraint on seller share of impressions	Operationalizes fairness: Gini=0 equal, Gini=1 fully concentrated
Delayed 30-day attribution window	Last relevant impression within configurable window	Models multi-session purchase decisions; reduces last-click noise
Offline A/B simulation with HOLD_SIMULATED	Temporal holdout; structured decision artifact per run	HOLD_SIMULATED is the key product decision — automated ship would be wrong

What Was Deliberately Not Built

Real-time serving. No real-time ranking inference endpoint was built. PulseRank is an evaluation and governance platform — the question it answers is whether a ranker is good enough to ship, not how to serve it at millisecond latency. Serving infrastructure is the deployment layer, not the evaluation layer.

Contextual bandits or reinforcement learning. Explicit scope boundary. Contextual bandits would require a live feedback loop — actions, observations, reward signals — that does not exist in an offline evaluation setting. Adding bandits without live traffic is simulation theater; it would not be a defensible claim.

Live online A/B testing. No real traffic, no live experiment. The offline A/B simulation is explicitly labeled `HOLD_SIMULATED` to distinguish it from a live result. This is not a gap — it is intellectual honesty. The simulation logic is real; the traffic is synthetic.

Real seller and buyer users. All data is synthetic: 650 items, 80 sellers, 4,132 sessions seeded reproducibly. The mathematical relationships are correct; the population is not real. Every claim is bounded by this truth boundary.

The Key Product Decision: `HOLD_SIMULATED`

`HOLD_SIMULATED` is not a failure — it is the correct decision. IPS-corrected metrics show improvement, but the relative lift in the holdout segment does not clear the significance threshold at the current simulated traffic volume. The system correctly withheld a ship recommendation. Automated ship would have been wrong.

A product that always ships when IPS metrics improve is not a governance framework — it is a rubber stamp. The structured decision artifact that produces `HOLD_SIMULATED` includes the effect size, the significance threshold, and the recommended action. That artifact is the product. A future run with more traffic might produce `SHIP_SIMULATED` — and the decision logic would be the same.

3

Failure Mode Addressed

Is This Ranking Model Actually Better, or Is the Evaluation Lying?

The position-bias trap is a specific and common failure mode in marketplace ranking: evaluation and training data are both corrupted by the same display decisions, making the corruption invisible to standard metrics. Teams operating on biased metrics cannot tell whether a new ranker is genuinely better or whether it has learned to exploit position artifacts in the training data.

PulseRank addresses the position-bias trap by treating it as a missing-data problem. Every impression is logged with its display rank at serve time. A position-click curve is estimated from logged data, modeling the probability that a user clicks an item at position k given only that it was shown there. IPS weights correct for this: $\text{click} \times (1/\text{propensity}(\text{rank}))$. The result: NDCG@10 from 0.134 to 0.522. The display layer's decisions are no longer charged to the model.

The IPS Correction: Mathematical Grounding

The Inverse Propensity Score (IPS) correction is grounded in the Rosenbaum-Rubin framework for causal inference from observational data. In the marketplace context:

$$\text{IPS-NDCG} = \sum_i (\text{rel}_i / \text{propensity}(\text{rank}_i)) / \text{ideal_DCG}$$

Where $\text{propensity}(\text{rank}_i)$ is the probability that item i was clicked given only its display position — estimated from the historical impression log. Items shown at lower ranks are upweighted; items shown at rank 1 are downweighted. IPS weights are clipped at $\text{max_weight}=5.0$ to control variance from low-propensity (low-rank) impressions.

The Retrieval Ablation: Each Component Earns Its Place

Strategy	NDCG@10 (biased)	NDCG@10 (IPS corrected)	Note
Popularity only	0.134	0.318	Baseline — bestseller bias
Content-based only	0.098	0.401	Better tail, weaker popular
Hybrid (no rerank)	0.134	0.491	Coverage improves
Hybrid + MMR + Gini	0.134	0.522	Full system — governance adds fairness

4 What Success Looks Like

Product Metrics: What Would Move If This Works

Success for PulseRank is measurable in marketplace health outcomes. These are the metrics a marketplace operator would track to determine whether the ranking system is working — not model benchmarks.

SELLER GINI COEFFICIENT	CATALOG COVERAGE AT TOP	CONVERSION ATTRIBUTION	A/B DECISION QUALITY
Impression diversity	Coverage@10 = 0.226	Delayed window accu	Ship decisions that hold

Seller Gini Coefficient

The Gini coefficient measures the concentration of impressions across sellers. Gini=0 means perfectly equal distribution; Gini=1 means one seller receives all impressions. PulseRank's reranking system targets a governance-constrained Gini — after rerank: 0.582. Success means this number trends down over time without degrading relevance scores, indicating that diversity

and quality are improving together.

Catalog Coverage at Top-10

The percentage of catalog items that appear in at least one session's top-10 results. Coverage@10 = 0.226 — 147 of 650 items surface in top-10 across 4,132 sessions. A pure relevance ranker would concentrate top-10 on the same high-popularity items across all sessions. Success means coverage increases without a corresponding drop in purchase conversion — confirming that surfaced items were genuinely relevant, not arbitrary.

Conversion Attribution Accuracy

The difference in conversion rate attributed by the delayed-window method versus naive last-click. Marketplace purchase decisions frequently span multiple sessions — a buyer discovers an item, returns days later, and converts. Naive last-click misattributes these conversions, corrupting the training signal for the ranker. The delayed window (configurable, default 30 days) correctly captures these multi-session patterns.

A/B Decision Quality

The percentage of ship decisions — SHIP_SIMULATED verdicts — that would hold up post-launch if the experiment were run live. HOLD_SIMULATED is not a failure metric; it is the correct governance outcome when lift does not clear significance. Success means that SHIP_SIMULATED decisions are only issued when the evidence genuinely warrants them — not when IPS metrics look directionally positive.

Key Results Summary

Metric	Value	Significance
Sessions / Impressions	4,132 / 41,320	Evaluation corpus size
Naive NDCG@10 (biased labels)	0.134	Baseline — position bias baked in
IPS-corrected NDCG@10	0.522	True model quality (+289%)
Hybrid Recall@100	0.690	69% of purchases in candidate set
Seller Gini (post-rerank)	0.582	Governance constraint active
Catalog Coverage@10	0.226	147 / 650 items in top-10
Offline A/B decision	HOLD_SIMULATED	Lift exists; significance not cleared

5 One Honest Gap

PulseRank's IPS Correction Assumes a Stationary Position-Click Curve

The IPS correction assumes that rank 1 always gets roughly the same click uplift — that the position-click curve is stationary across query types, devices, time of day, and user segments. In practice, position bias varies significantly across these dimensions. A single global propensity curve is a simplification.

The gap is real. Position bias is not homogeneous across the marketplace. Consider three dimensions where the propensity curve shifts:

- **Query type.** Navigational queries (user searching for a specific seller) show much weaker position bias than exploratory queries (user browsing a category). A single propensity curve averaged across both query types underestimates bias for exploratory and overestimates it for navigational.
- **Device.** Mobile interfaces show stronger position bias than desktop — the visible fold is smaller, fewer items appear without scrolling. A global curve trained on a mixed device corpus applies the average to contexts where the true propensity is meaningfully different.
- **User segment.** First-time visitors show stronger position bias than repeat buyers who know the catalog. A segment-level propensity curve would more accurately model the click process for each population.

Why Segment-Level Correction Was Not Built

Estimating segment-level propensity curves requires significantly more data per segment. With 41,320 impressions across 4,132 sessions, the global curve is well-estimated. Splitting by device (2 levels) x query type (4+ categories) x user segment (several buckets) reduces each cell to a few hundred impressions — insufficient for reliable propensity estimation. Underestimated propensity in low-data segments would introduce variance that makes the correction noisier than the bias it removes. The cure would be worse than the disease.

Global Correction Is the Right Default

Global IPS correction is the right default for a production ranking system. It removes the first-order position bias — the largest and most systematic distortion in the evaluation signal. Segment-level correction is a research direction that becomes viable as impression volume scales and as the confidence intervals on segment-level propensity estimates narrow.

The deliberate scope boundary here is intellectually honest: the global curve is a simplification, but it is a well-motivated one. A team building on PulseRank would add segment-level propensity estimation as traffic grows — not as a Day 1 requirement, but as a natural evolution of the evaluation infrastructure.

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